Data Augmentation for Machine Learning

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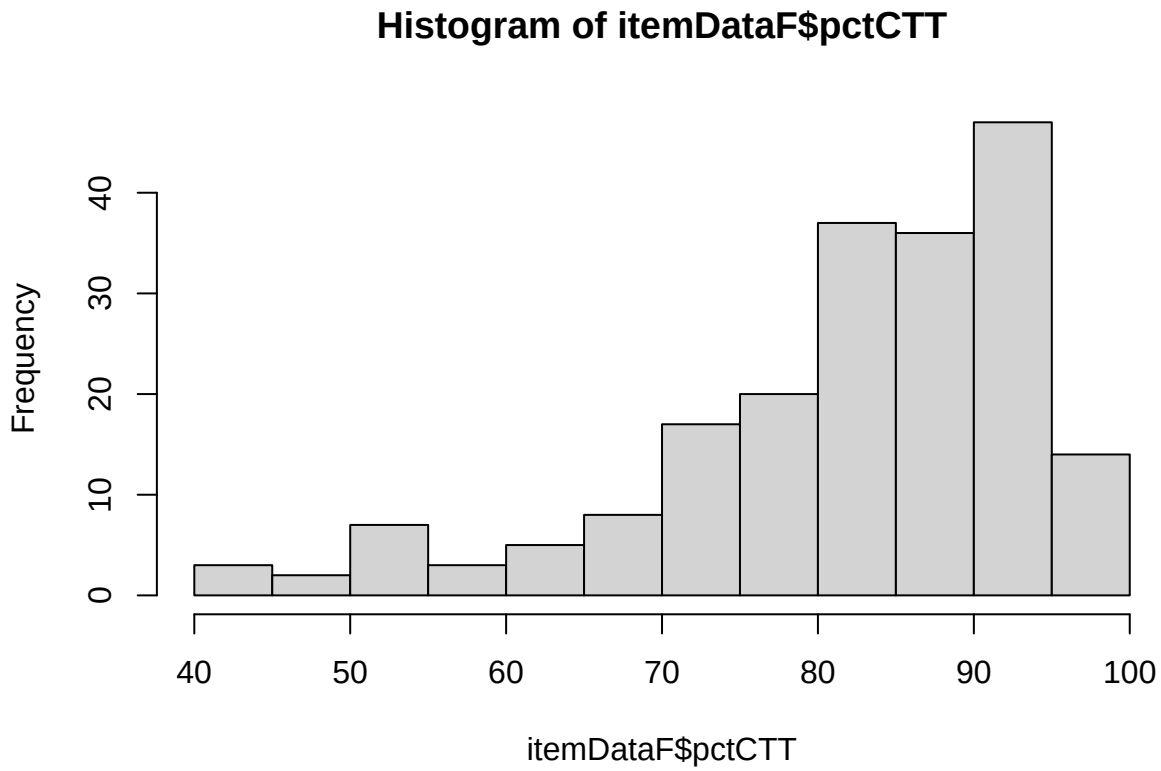
Abstract

Purpose: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.

Keywords: data augmentation, SmoteR, Catalytic Prior

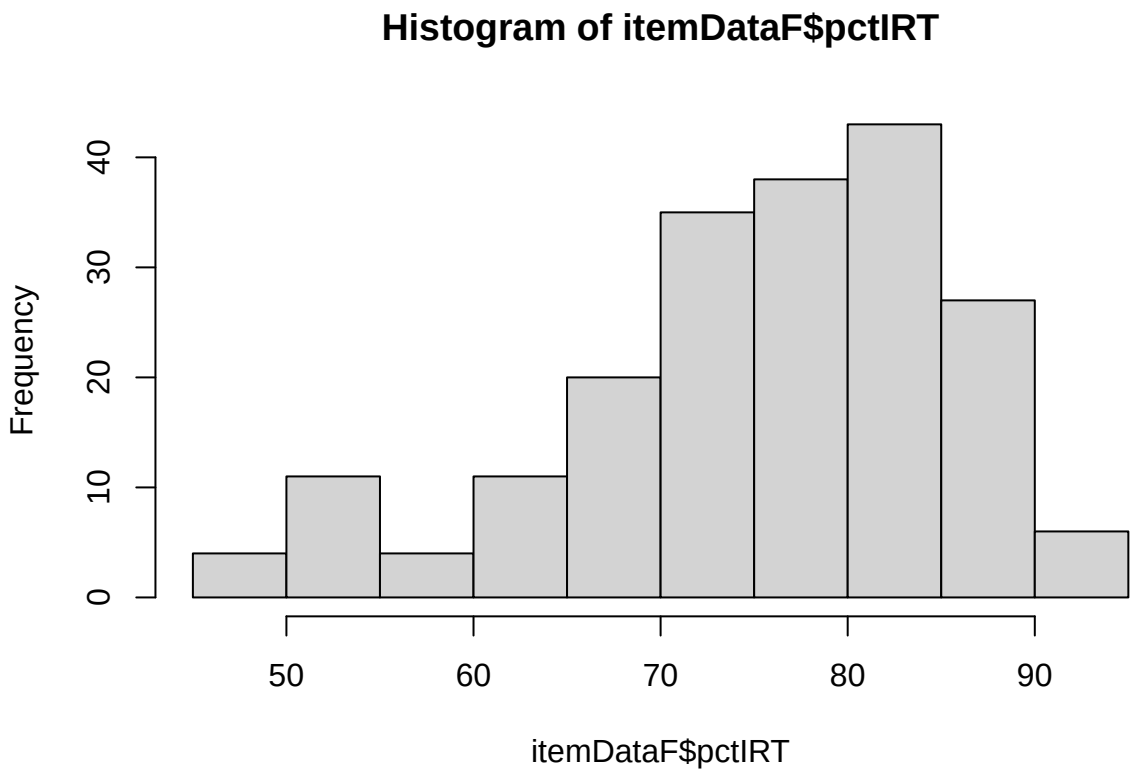
JEL Classification: D8 , H51

MSC Classification: 35A01 , 65L10




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## $counts
##  [1]  3  2  7  3  5  8 17 20 37 36 47 14
##
## $density
##  [1] 0.003015075 0.002010050 0.007035176 0.003015075 0.005025126 0.008040201
##  [7] 0.017085427 0.020100503 0.037185930 0.036180905 0.047236181 0.014070352
##
## $mids
##  [1] 42.5 47.5 52.5 57.5 62.5 67.5 72.5 77.5 82.5 87.5 92.5 97.5
##
## $xname
## [1] "itemDataF$pctCTT"
##
## $equidist
```

```
## [1] TRUE
##
## attr(,"class")
## [1] "histogram"
```




```
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##  [1] 45 50 55 60 65 70 75 80 85 90 95
##
## $counts
##  [1]  4 11  4 11 20 35 38 43 27  6
##
## $density
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##  [7] 0.038190955 0.043216080 0.027135678 0.006030151
##
## $mids
##  [1] 47.5 52.5 57.5 62.5 67.5 72.5 77.5 82.5 87.5 92.5
##
```

```
## $xname
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##
## $equidist
## [1] TRUE
##
## attr(,"class")
## [1] "histogram"
```

1 Introduction

The objective of data augmentation is to improve the performance of learning model prediction on the rare cases. If there are majority class, the model learn the data focusing on the majority class, which lead to lower the performance of predicting on rare cases.

The primary objective of data augmentation is to generate "good" synthetic data which are rare cases. In this context, "good" synthetic data effectively capture the underlying relationship between predictor and target variables within training set. The ultimate goal is to train a model that can accurately predict target variables from the predictor variable in an unseen test set. To achieve this, we explored several data augmentation techniques designed to create well-connected predictor-target pairs. The methods investigated include random oversampling, Gaussian Noise, SMOTER, SMOTGN, Importance Sampling, Multiple Imputation, and the use of a catalytic prior.

2 Data Augmentation

2.1 Random Oversampling

Replicate original data including target variable and predictors within the threshold.

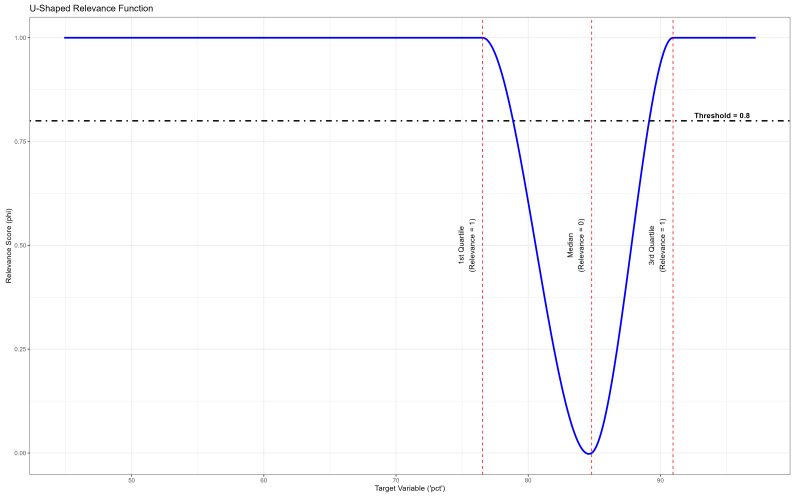


Fig. 1 Relevance function and threshold for CTT

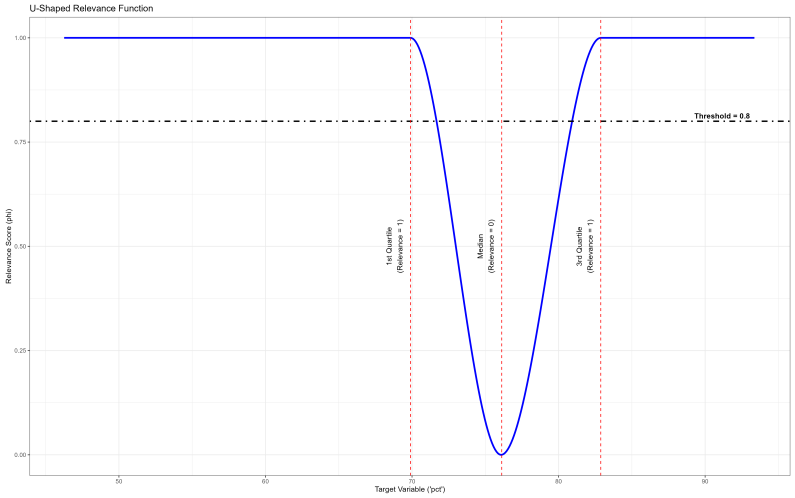


Fig. 2 Relevance function and threshold for IRT

2.2 Gaussian Noise

Add an error sampled from Normal(0, perc) to a rare case. Perc is derived from the median of distances with the other rare cases from the rare value. It is applied to the predictors and target value.

2.3 SMOTE R

Conduct interpolation between one rare case and the other rare case. The other rare case is randomly selected within the k-nearest neighbors among the rare cases. Between the two rare cases, interpolation is applied. This is applied to predictor and target variable.

2.4 SMOGN

This is combination of Smote R and Gaussian Noise. When the other rare case is within the safe zone which is close than the median of the distances derived from all the distance from the seed rare case. If the other rare case is not within the safe zone, then it is neglected and Gaussian Noise method is applied.

2.5 Importance Sampling

2.6 Multiple Imputation (DA-MI)

This three-step method creates a synthetic dataset that mimics the statistical properties of original, sensitive data. This allows researchers to replicate findings without compromising the privacy of individuals. 1. Generating pseudo-observations (w) by applying a masking model to the original data. This process intentionally adds statistical noise or otherwise perturbs the confidential values, effectively creating a masked version of the dataset that obscures the true, sensitive information. 2.Next, fit a joint imputation model by integrating two distinct parts: the masking model from the previous step and an association model. This association model is designed to capture the underlying statistical relationships that is derived from the analysis model applied with the variables in the original dataset. By combining these components, the joint model simultaneously learns both the data's true patterns and the specific nature of the noise that was introduced. 3. Finally, generate the synthetic data by drawing new values from the Bayesian posterior predictive distribution of the fitted joint model.

A key methodological distinction between the DA-MI framework and the Gaussian noise method is found in their respective approaches to synthesizing the dependent variable. The DA-MI approach generates these values from a formal statistical model, drawing new observations from the Bayesian posterior predictive distribution. Conversely, the Gaussian noise method employs a more direct perturbation technique; it creates synthetic values by adding random noise, sampled from a Gaussian distribution, directly to the original data points.

2.7 Catalytic Prior

To improve model stability and prevent overfitting, a dataset can be augmented with synthetic data generated from a simpler, more robust model. This technique works by using the parameter estimates from the simpler model to create new, artificial data points. This can be done even if the parameter estimates themselves are not statistically significant. The purpose of this synthetic data is not to find the "true" underlying model, but rather to increase the effective sample size for the more complex model. By expanding the dataset, this process stabilizes the complex model and acts

as a form of regularization, preventing overfitting by shrinking the complex model's parameters toward the simpler, often intercept-only, baseline.

The primary difference between the DA-MI (Data Augmentation with Multiple Imputation) and Catalytic prior methods for generating synthetic data lies in the mechanism for creating values for independent variables. The Catalytic prior method generates values by drawing them randomly from the possible range of an independent variable. This approach is not constrained by the combinations observed in the original data, which can result in the creation of novel data patterns not present in the initial dataset. Consequently, this method is well-suited for exploring the potential data space and testing model robustness against new scenarios. In contrast, the DA-MI method is designed to generate synthetic values that are as similar as possible to the original data. Its fundamental purpose is to support the faithful replication of research; by creating a dataset that closely mirrors the original's characteristics and distributions, DA-MI facilitates analysis that yield results consistent with the original study without disclosing sensitive information. Therefore, the choice between these methods is contingent upon the research goal: the Catalytic prior is preferable for exploration and pattern discovery, whereas DA-MI is the appropriate choice for validation and replication.

3 Machine Learning

Ridge, Lasso, SVR, KNN, Logistic Regression, Neural networks are sensitive to the scale of features. Tree-based models like Random Forest, XGBoost separate data not based on the size, but the order or separation point.

3.1 Ridge and Rasso

3.2 SVR

This model calculate the distance between the data points, which makes the model become sensitive to the scale of the features.

3.3 Random Forest

3.4 XGBoost

3.5 Cubist

Q. In catalytic prior, when generating y value, the value can over 100. Do we need to constrain them to 100?

4 Results

This is result section.

Table 1 Frequency table by data augmentation method

X	Var	X0.50	X50.60	X60.70	X70.80	X80.90	X90.100	X100.497	Total
train_random_NA	pctCTT	2	8	10	28	51	40	0	139
train_random_RO	pctCTT	3	29	12	28	51	40	0	163
train_random_GN	pctCTT	4	16	12	28	51	40	0	151
train_random_SR	pctCTT	4	18	10	28	51	40	0	151
train_random_SG	pctCTT	4	16	12	28	51	40	0	151
train_random_IS	pctCTT	5	14	13	28	51	40	0	151
train_random_MI	pctCTT	4	16	20	56	102	80	0	278
train_random_CT	pctCTT	4	10	10	28	51	41	22	166
train_25g_NA	pctCTT	5	7	12	29	64	57	0	174
train_25g_RO	pctCTT	15	24	23	29	64	57	0	212
train_25g_GN	pctCTT	10	14	19	29	64	57	0	193
train_25g_SR	pctCTT	13	13	17	29	64	57	0	193
train_25g_SG	pctCTT	11	13	19	29	64	57	0	193
train_25g_IS	pctCTT	11	15	17	29	64	57	0	193
train_25g_MI	pctCTT	10	14	24	58	128	114	0	348
train_25g_CT	pctCTT	14	8	16	30	65	58	8	199
train_random_NA1	pctIRT	1	11	23	54	47	3	0	139
train_random_RO1	pctIRT	4	32	23	54	47	3	0	163
train_random_GN1	pctIRT	2	22	23	54	47	3	0	151
train_random_SR1	pctIRT	2	22	23	54	47	3	0	151
train_random_SG1	pctIRT	2	22	23	54	47	3	0	151
train_random_IS1	pctIRT	0	24	23	54	47	3	0	151
train_random_MI1	pctIRT	2	22	46	108	94	6	0	278
train_random_CT1	pctIRT	4	11	24	55	48	5	19	166
train_25g_NA1	pctIRT	4	12	27	61	64	6	0	174
train_25g_RO1	pctIRT	10	36	27	61	64	6	0	204
train_25g_GN1	pctIRT	8	23	27	61	64	6	0	189
train_25g_SR1	pctIRT	11	20	27	61	64	6	0	189
train_25g_SG1	pctIRT	7	24	27	61	64	6	0	189
train_25g_IS1	pctIRT	9	22	27	61	64	6	0	189
train_25g_MI1	pctIRT	8	24	54	122	128	12	0	348
train_25g_CT1	pctIRT	16	13	28	62	66	10	5	200

Table 2 Evaluation metric by data augmentation method

	RMSE		MAE		R-squared		MAPE	
Aug.	Mean	SD	Mean	SD	Mean	SD	Mean	SD
NA	9.322	1.128	6.944	0.549	0.387	0.067	10.320	0.902
SR	9.469	1.099	7.029	0.543	0.346	0.065	10.216	0.763
IS	9.591	1.205	7.061	0.680	0.332	0.095	10.293	1.028
MI	9.674	1.087	7.472	0.587	0.276	0.102	10.752	0.946
SG	9.715	1.317	7.191	0.738	0.326	0.089	10.403	0.926
GN	9.854	1.483	7.271	0.827	0.311	0.113	10.556	1.078
RO	9.974	1.207	7.504	0.751	0.295	0.085	10.688	1.007
CT	11.615	3.505	8.469	2.348	0.280	0.106	12.081	3.015



Fig. 3 RMSE across data augmentation methods for Lasso

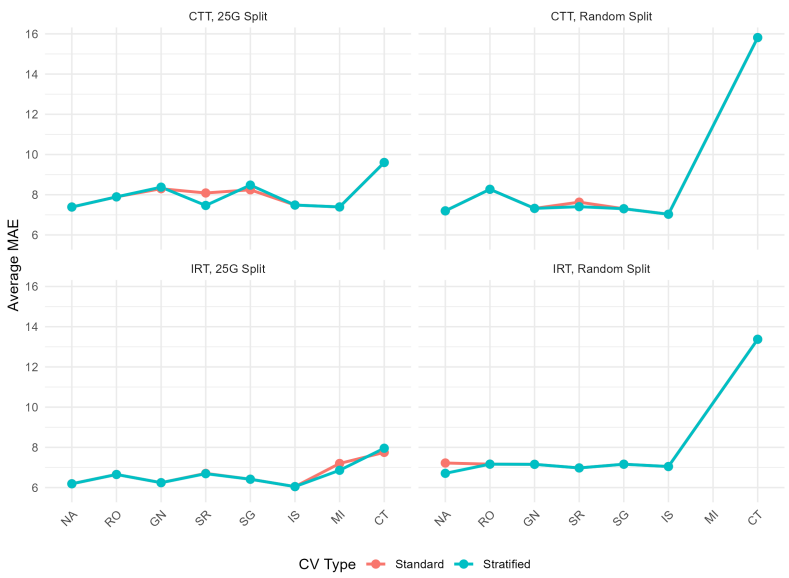


Fig. 4 MAE across data augmentation methods for Lasso

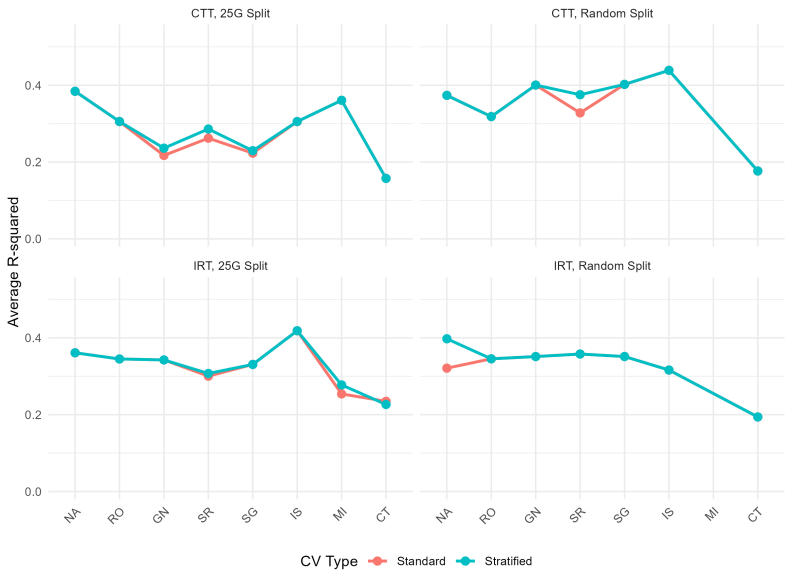


Fig. 5 R-squared across data augmentation methods for Lasso

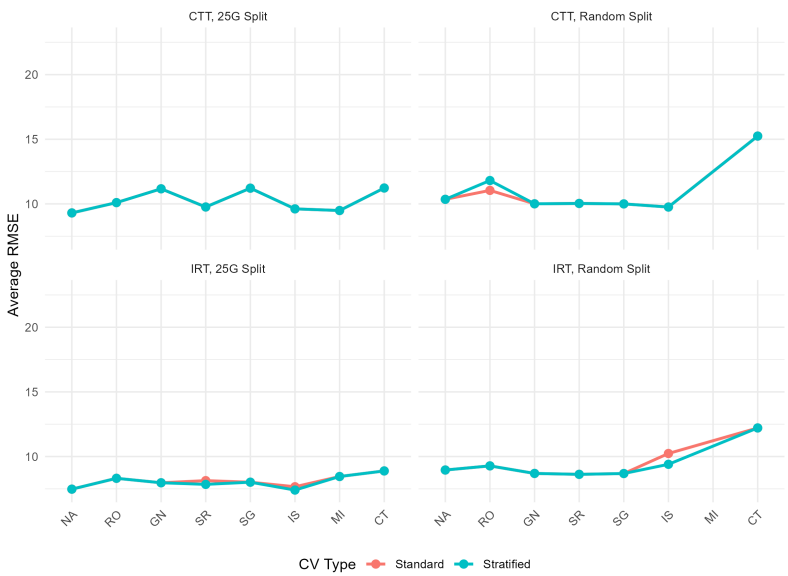


Fig. 6 RMSE across data augmentation methods for Ridge

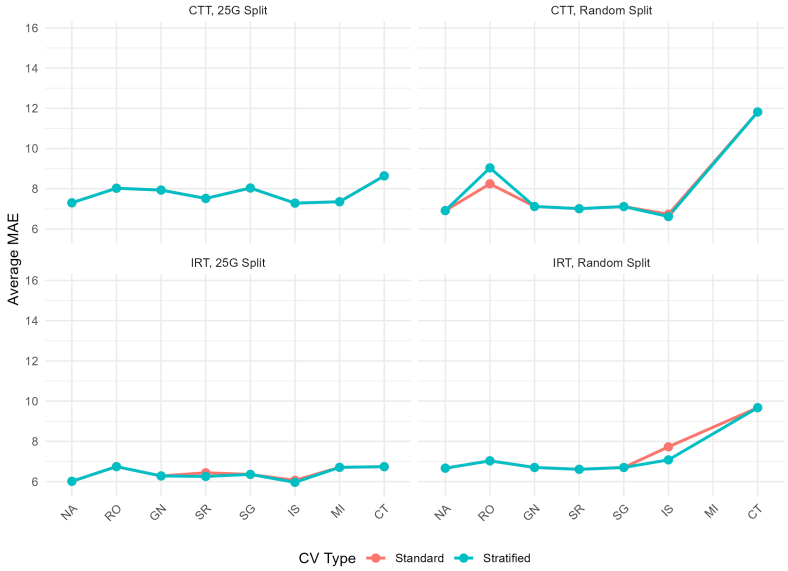


Fig. 7 MAE across data augmentation methods for Ridge

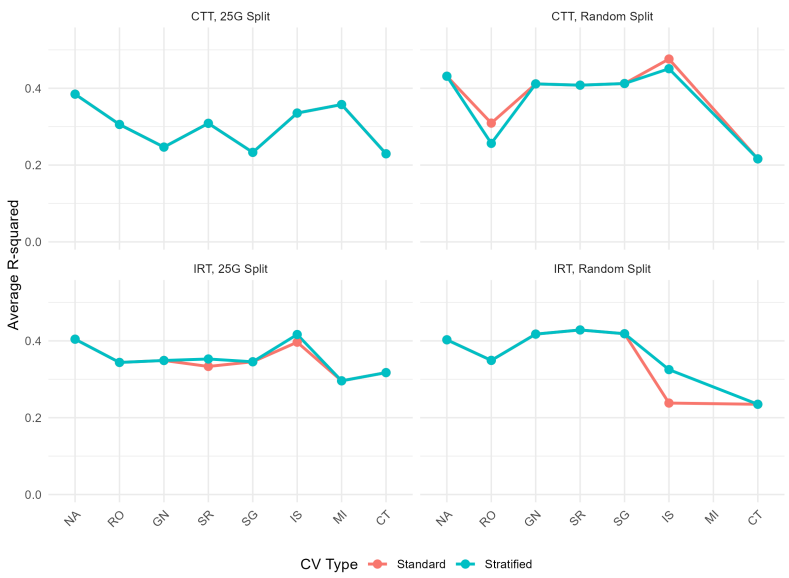


Fig. 8 R-squared across data augmentation methods for Ridge



Fig. 9 RMSE across data augmentation methods for Random Forest



Fig. 10 MAE across data augmentation methods for Random Forest

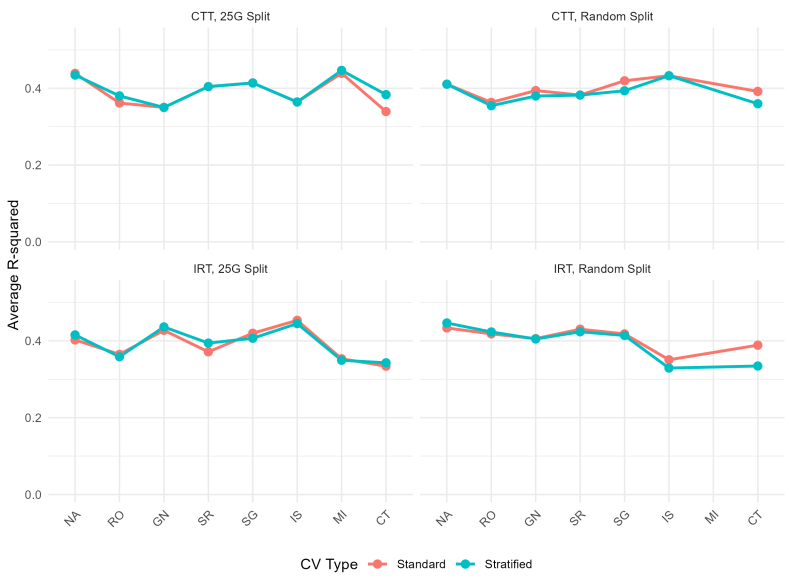


Fig. 11 R-squared across data augmentation methods for Random Forest



Fig. 12 RMSE across data augmentation methods for XGBoost



Fig. 13 MAE across data augmentation methods for XGBoost

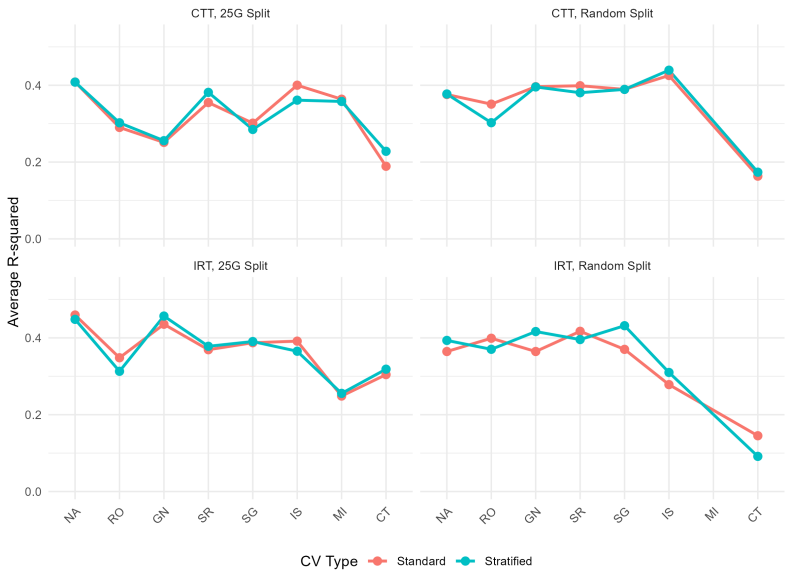


Fig. 14 R-squared across data augmentation methods for XGBoost

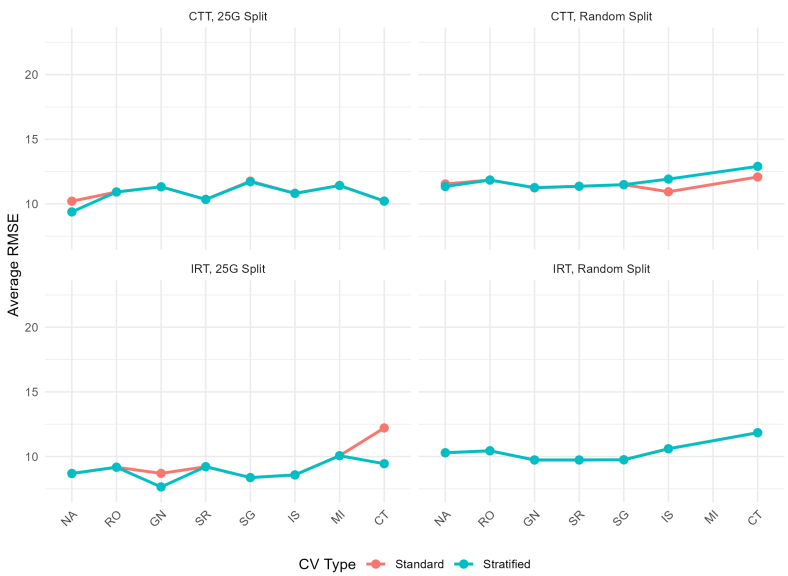


Fig. 15 RMSE across data augmentation methods for Cubist



Fig. 16 MAE across data augmentation methods for Cubist



Fig. 17 R-squared across data augmentation methods for Cubist

Overall learning model: GN, SR, SG performed well. Lasso, Ridge (Regularized linear regression model): GN, SG, IS, MI Random Forest, XGBoost, Cubist: GN, SG, CT

Random Forest and XGBoost are tree-based method. , Cubist is mixture of tree-based and linear regression. Random Forest build hundreds of decision trees independently of one another. XGBoost builds trees sequentially. The first tree makes a prediction, and the second tree is then trained based on the errors (residuals) of the first tree. The third tree is trained based on the error of second tree. Therefore, new tree focused on the hardest-t-predict cases. This sequential error-correcting approach makes it estremely powerful and often results in higher accuracy than Random Forest.

Except RO and IS, all results of augmentation methods outperformed than the NA. SR was always the best across all conditions.

MAE also showed the similar pattern withh the RMSE except CTT and Random split showing the higher MAE for CT than NA.

MAPE also showed the similar pattern with the MAE.

In R-squared, except RO and IS, all methods showed matter perforamnce than the NA, except CTT and Random split condition.

Next Step Various tuning: threshold, oversampling ratio, the number of synthetic data

Add Measurement criteria focusing on rare cases: F1 score, RMSE_rare

F1 score: 2x(precision x recall)/(precision+recall) Precision: TP/(TP + FP) Recall: TP/(TP + FN) RMSE for rare cases based on threshold